Environment Design

A group of 10 self-interested agents sharing a common resource. Each round, the agents update their harvest rate based on the expected reward. Concretely, the agents choose whether to harvest gently or aggressively from their "fair-share" ("fair-share" is the total available resource divided by the number of agents). I model the reward for each agent as the "revenue" they earn from the harvest minus the "cost" of harvest. The revenue is modeled to be linear in harvest whereas the cost is convex. This leads us to the reward equation:

$$reward = (1 - tax) * r * h - c * h^2 + (redistributed\ tax),$$

where:

- tax is the applied tax rate
- r is a constant that scales the "revenue"
- c is a constant that scales the "cost"

Following the problem setup, each agent is also assigned a greed-level, but I model the greed as inverse softmax temperature. A higher greed level more strongly prefers a higher reward.

Resource re-growth is modeled as logistic growth:

$$H_{t+1} = H_t - (harvest)_t + g * H_t * (1 - H_t),$$

where:

- H is the field health
- t is the current round
- $(harvest)_t$ is the total amount harvested in round t
- g is the re-growth constant

Policymaker

The rule-based policymaker makes decisions based on a "target field health" and a "minimum acceptable reward". The update to the tax-rate is calculated like this:

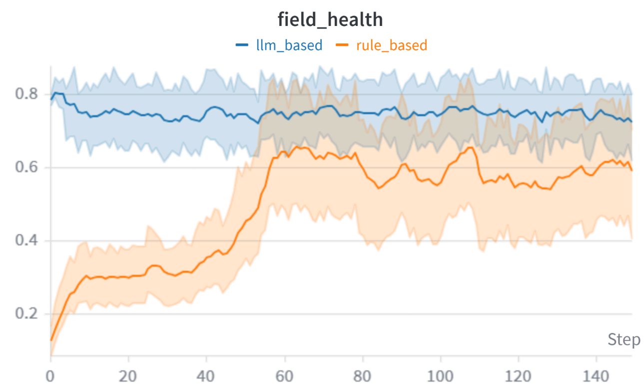
$$delta = w_h * max(0, TH - H) - w_r * max(0, MR - avg_r),$$

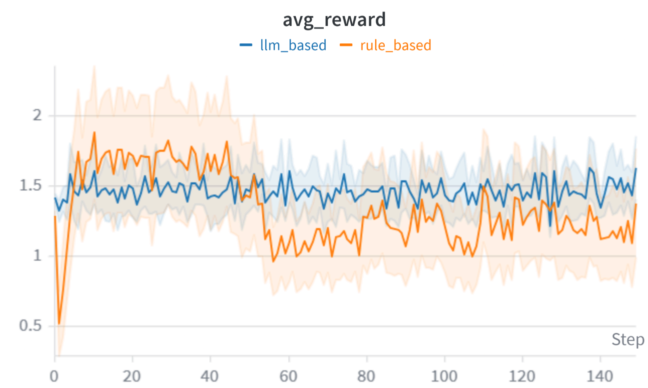
where:

- TH is the target health
- MR is the minimum acceptable reward
- H is the field health
- avg_r is the average agent reward
- w_h and w_r are constants that decide how these two updates are linearly combined

For the LLM policymaker, I use `qwen2.5:1.5b` locally via Ollama resulting in **0 inference cost**.

Results (across 5 seeds)





For average field health in the last 50 rounds, the LLM achieves 0.748 on average, outperforming the rule-based policymaker's 0.588. Check out the full breakdown here.

Across all my experiments, the LLMs tended to stick to whatever tax-rate they happened to pick first. In the current formulation, the LLMs also don't have a sense for what is a "decent" agent reward (as the order of magnitude for agent rewards is arbitrary) and therefore generally perform worse than the rule-based method on this front. I suspect both these issues can be mitigated by using a more capable LLM.

Conclusion

The LLM policymaker performs better on average by avoiding catastrophic collapse, but it does not demonstrate a stable or interpretable strategy. Its behavior is largely reactive and exhibits anchoring, suggesting that with limited context and model capacity, it is not reliably performing sequential decision-making.

Thing I'd like to change

I would more fundamentally rethink how **agent rewards and engagement** are modeled. Currently, agents always participate and optimize a fixed reward function. In practice, poorly designed incentives should affect participation — agents may reduce effort, change behavior, or drop out entirely. Introducing dynamic participation or evolving preferences would create a more realistic environment and force the policymaker to balance not just resource sustainability but also continued engagement.

One question the task raised

This task made me think about the limits of **in-context learning**. To what extent is the LLM actually adapting its behavior over time versus simply reacting to recent observations? The anchoring behavior suggests that short-context prompting may not be sufficient for true iterative reasoning in environments with delayed and nonlinear dynamics.